Transfer API - Assignment

Overview

Backend API for a simple transaction ledger system.

Assumptions

- All monetary values are stored in cents to avoid rounding issues, with a maximum of two decimal places.
- New accounts start with a zero balance and receive a unique ID.
- Debits are allowed only when sufficient funds are available.
- Transfers must succeed completely or have no effect, ensuring balances remain consistent.
- Transactions cannot be modified once recorded.
- Pagination is supported, with a default limit of 10 and a maximum of 100 records per request.
- SQLite is used as the database, with transactions to maintain data consistency.
- Authentication is outside the scope of this assignment.

Version information

Version : 0.0.1

License information

Terms of service : /tos

Paths

Create a new account

```
POST /api/accounts
```

Parameters

Type	Name	Schema
Body	**body** *required*	body

body

Name	Description	Schema
name *required*	**Example** : `"Savings Account"`	string

Responses

HTTP Code	Description	Schema
201	Account created successfully	No Content
400	Invalid request or initial balance supplied	No Content
415	Content-Type must be application/json	No Content

Consumes

- `application/json`

Produces

- `application/json`

Tags

- Accounts

Retrieve accounts with pagination

```
GET /api/accounts
```

Parameters

Type	Name	Schema	Default
Query	**limit** *optional*	integer	`10`
Query	**page** *optional*	integer	`1`

Responses

HTTP Code	Description	Schema
200	Paginated account list	Response 200
400	Invalid pagination parameters	No Content

Response 200

Name	Schema
accounts *optional*	< accounts > array
pagination *optional*	pagination

accounts

Name	Description	Schema
balance *optional*	**Example** : `"100.00"`	string
created_at *optional*		string
id *optional*		string (uuid)
name *optional*		string

pagination

Name	Schema
limit *optional*	integer
page *optional*	integer
total_pages *optional*	integer
total_records *optional*	integer

Tags

- Accounts

Retrieve an account balance

```
GET /api/accounts/{account_id}/balance
```

Parameters

Type	Name	Schema
Path	**account_id** *required*	string (uuid)

Responses

HTTP Code	Description	Schema
200	Account balance retrieved	Response 200
404	Account not found	No Content

Response 200

Name	Description	Schema
account_id *optional*		string (uuid)
balance *optional*	**Example** : `"250.00"`	string

Tags

- Accounts

Create a credit or debit transaction

```
POST /api/accounts/{account_id}/transactions
```

Parameters

Type	Name	Schema
Path	**account_id** *required*	string (uuid)
Body	**body** *required*	body

body

Name	Description	Schema
amount *required*	**Minimum value (exclusive)** : `0` **Example** : `100.5`	number

Name	Description	Schema
description *required*	**Example** : `"Initial deposit"`	string
reference_id *required*	**Example** : `"TXN-001"`	string
type *required*		enum (credit, debit)

Responses

HTTP Code	Description	Schema
201	Transaction created successfully	No Content
400	Invalid transaction data	No Content
404	Account not found	No Content
409	Duplicate transaction reference	No Content
415	Content-Type must be application/json	No Content
422	Insufficient balance or balance limit exceeded	No Content
500	Database error	No Content

Consumes

- `application/json`

Tags

- Transactions

Retrieve paginated transaction history

```
GET /api/accounts/{account_id}/transactions
```

Parameters

Type	Name	Schema	Default
Path	**account_id** *required*	string (uuid)	
Query	**limit** *optional*	integer	`10`

Type	Name	Schema	Default
Query	**page** *optional*	integer	1
Query	**type** *optional*	enum (credit, debit)	

Responses

HTTP Code	Description	Schema
200	Transaction history retrieved	Response 200
400	Invalid pagination or transaction type	No Content
404	Account not found	No Content

Response 200

Name	Schema
data *optional*	< data > array
pagination *optional*	pagination

data

Name	Description	Schema
account_id *optional*		string
amount *optional*	**Example** : `"100.50"`	string
created_at *optional*		string
description *optional*		string
id *optional*		string
reference_id *optional*		string
type *optional*		enum (credit, debit)

pagination

Name	Schema
limit *optional*	integer
page *optional*	integer
total_pages *optional*	integer
total_records *optional*	integer

Tags

- Transactions

Transfer funds between two accounts atomically

```
POST /api/transfers
```

Parameters

Type	Name	Schema
Body	**body** *required*	body

body

Name	Description	Schema
amount *required*	**Minimum value (exclusive)** : `0` **Example** : `30.0`	number
description *required*	**Example** : `"Payment for services"`	string
destination_account_id *required*	**Example** : `"22222222-2222-4222-8222-222222222222"`	string (uuid)
reference_id *required*	**Example** : `"TRANSFER-001"`	string
source_account_id *required*	**Example** : `"11111111-1111-4111-8111-111111111111"`	string (uuid)

Responses

HTTP Code	Description	Schema
201	Transfer completed successfully	Response 201
400	Invalid request or identical source and destination	No Content
404	Source or destination account not found	No Content
409	Transfer reference already used	No Content
415	Content-Type must be application/json	No Content
422	Insufficient source balance or destination balance limit exceeded	No Content
500	Database error or unexpected transfer failure	No Content

Response 201

Name	Description	Schema
amount *optional*	**Example** : `"30.00"`	string
created_at *optional*		string
description *optional*		string
destination *optional*		destination
reference_id *optional*		string
source *optional*		source

destination

Name	Schema
account_id *optional*	string
balance *optional*	string
transaction *optional*	object

source

Name	Schema
account_id *optional*	string
balance *optional*	string
transaction *optional*	object

Consumes

- `application/json`

Produces

- `application/json`

Tags

- Transfers